1/ KYC ENGINE

هذا هو Customer Identity & Risk Profil
ماشي غير:
"هل عندنا وثائق العميل؟"
بل يبني Customer Risk Profile
البيانات:

- **كي تدخل Client واحد**

مثلا:

text

CUSTOMER PROFILE

Customer ID: C-001245
Status: ACTIVE

KYC STATUS: 🟢 VERIFIED
RISK LEVEL: 🟠 MEDIUM
KYC SCORE: 58/100

Last KYC Review:
06/10/2026

Next Review:

06/10/2027

وتحتها Tabs:

```text
[Identity]
[Business]
[Financial]
[UBO]
[Documents]
[Risk]
[Behavior]
[Changes]
[History]
```

● Identity

```text
IDENTITY
```

Customer ID
Full Name
Date of Birth
Nationality
Country of Residence
Address
Phone
Email

Identification Document
Document Type
Document Number
Issue Date
Expiry Date
Verification Status

النظام يعطي:

```text
Identity Verification: 🟢 VERIFIED
Document Status: 🟢 VALID
```

● Business / Professional Profile

حسب نوع العميل. (أدوما ال variables ta3 l customers)

Individual

text

Profession
Employer
Employment Status
Sector
Country of Employment
Expected Income
Purpose of Account

Company

text

Legal Name
Legal Form
Registration Number
Business Activity
Sector

Country of Incorporation
Countries of Operation
Annual Turnover
Number of Employees
Purpose of Account

● Financial Profile

هنا مهم خلاه

text

FINANCIAL PROFILE

Declared Income
Declared Turnover
Source of Funds
Source of Wealth

Expected Monthly Inflow
Expected Monthly Outflow

Expected Transaction Volume
Expected Transaction Amount

Cash Activity
International Activity

لكن: هاذي معلومات KYC المصرح بها.
بعدها RiskVision يقارنها مع **actual data**.

● **RiskVision يبني:**

CUSTOMER BEHAVIORAL BASELINE
أدي رايحا تكون على حساب les variables li mektoubin koll ta3 PD, Fraud system, anomaly system

هاذي **ما يدخلهاش الموظف يدويا.**
historical transaction data. يبنيها من RiskVision

● KYC Risk Engine

نحسب Risk Score من عدة dimensions:

text

KYC RISK ENGINE

Identity Risk 10/100
Business Risk 35/100

```
Financial Risk          48/100
Geographic Risk          20/100
Product Risk            30/100
Channel Risk            42/100
Behavioral Risk          67/100
KYC Data Quality          5/100
                    ──────────
                        58/100
```

ثم:

```text
0–29    LOW
30–59   MEDIUM
60–79   HIGH
80–100  CRITICAL
```

والـweights ما تكونش بالضرورة ثابتة a bro, الـCompliance يقدر يضبطها حسب سياسة البنك.

● أهم جزء: KYC Change Detection

هنا RiskVision يراقب **التغيرات**.

مثلاً:

text

KYC PROFILE CHANGE DETECTED

Customer: C-001245

Changes detected:

✓ Contact information changed
✓ New device detected
✓ Transaction behavior changed
✓ New beneficiary pattern
✓ Transaction volume increased

Change Score: 76/100

Status:
🟠 REVIEW REQUIRED

بصح ما نقولوش **Fraud**.

Dk nzid nfahmek akter

A — Declared KYC

واش صرح به العميل:

Activity = Import/Export

Monthly Turnover = 2M DA

B — Registered KYC

واش موجود في البيانات الرسمية/البنكية المتاحة:

RC Activity = Import/Export

C — Observed Profile

واش RiskVision يشوف فعليا من العمليات:

Actual Turnover = 20M DA

Main counterparties = Construction companies

Countries = China / UAE / Turkey

● هنا يجي Dynamic KYC Engine

RiskVision يدير comparison بشكل مستمر:

```text
OLD KYC
  ↓
CURRENT DATA
  ↓
OBSERVED BEHAVIOR
```

↓

COMPARE

↓

KYC CHANGE DETECTION

مثلا:

Variable	KYC	Current	Result
Activity	Import/ Export	Import/ Export	🟢
Turnover	2M	20M	🔴 Major Change
Transactions	30	160	🟠 Change
Countries	2	6	🟠 Change
Beneficiaries	10	35	🟠 Change

النظام يخرج:

🟠 KYC PROFILE CHANGE DETECTED

وليس:

🔴 FRAUD

● كيفاش يعرف Activity تبدلت؟

هنا دير **Activity Classification Engine**.

ما تعتمدش على transaction واحدة.

لمدة معينة، مثلا transactions يجمع RiskVision

15/30/45 يوم، ويحلل:

- نوع العمليات
- counterparties
- transaction descriptions
- suppliers/customers
- countries
- amounts
- frequency
- incoming/outgoing flows

ثم يحدد:

```
Declared Activity
      ↓
Import / Export
```



مثلا:

```text
Declared Activity:
IMPORT / EXPORT

Observed behavior over 90 days:

Construction-related counterparties = 64%
Wholesale payments = 21%
Import/export = 15%

→ Observed Activity:
Construction / Wholesale

→ Activity Consistency = LOW
```

فتطلع:

🟠 ACTIVITY MISMATCH

● دير KYC Change Score

ماشي كل changement عندو نفس الخطورة.

مثلا:

```
KYC CHANGE SCORE
      0 ──────────────────────────────────────
100

Activity Change       +30
Turnover Change        +25
Country Change         +15
Beneficiary Change     +10
Transaction Pattern    +10
Legal Information      +10
```

ثم:

```text
0–29 🟢 Stable
30–59 🟡 Minor Change
60–79 🟠 Significant Change
80–100 🔴 Major Change
```

لكن هذا score خاص بـKYC Change، ماشي Fraud Score.

وبعدها تربطو بالـEWS

● مثال RiskVision

العميل كان:

```text
Customer: C-001245

Declared Activity:
Import / Export

Expected Turnover:
```

2,000,000 DA/month

Expected Countries:
Algeria / Tunisia

KYC Risk:
Medium

بعد 60 يوم:

text

Observed Turnover:
20,000,000 DA

Change:
+900%

New Countries:
China / UAE / Turkey

New Beneficiaries:
+27

Observed Activity:

Wholesale / Construction

RiskVision:

text



```
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      DYNAMIC KYC ALERT

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Customer: C-001245

🔴 KYC PROFILE CHANGE

Activity:
Import/Export
        ↓
Wholesale / Construction

Turnover:
2M DA
        ↓
20M DA
```

↑
+900%

Activity Consistency:
42/100

KYC Change Score:
78/100

Status:
SIGNIFICANT PROFILE CHANGE

ثم:

Action → Send to EWS

والـEWS يشوف الإشارات الأخرى.

إذا مثلاً:

text

KYC Change 78
Transaction Anomaly 82
New Beneficiary Risk 75
Geographic Risk 60

يقدر يخرج:

🔴 EWS HIGH RISK

لكن **Fraud Engine** وحده هو اللي يقرر إذا transaction معينة فيها fraud risk.

يعني باختصار: في RiskVision، الـKYC يكون Baseline + Continuous Monitoring + Change Detection. والـKYC ما يحكمش على العميل بأنه Fraud, هو ينتج KYC Signals، والـEWS يجمعها مع باقي الأنظمة باش يعطي الصورة الكاملة للـCustomer Risk.